

When Do Synthetic Hard Negatives Help? Scale-Dependent Contrastive Learning for IoT Intrusion Detection

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Abstract

Do synthetic hard negatives improve contrastive anomaly detection for IoT intrusion detection? We investigate using HNIDS (**H**ard-**N**egative **I**DS), a framework that synthesises stealth-controlled IoT attacks via closed-form perturbations with protocol compliance (Modbus, MQTT, CoAP), and fine-tunes Moirai [Woo et al.(2024)] with combined NLL and supervised contrastive loss. A five-condition ablation at three stealth levels with benign-calibrated thresholds isolates the contributions of hard negatives, contrastive learning, and fine-tuning against nine IDS baselines.

Our central finding is catastrophic forgetting of pre-trained features. At increased training scale, performance is non-monotonic: 500 samples improves over 200 (suggesting the smallest regime is under-trained), but degrades from 500 to 1000 samples as catastrophic forgetting erases pre-trained features. Freezing the encoder and training only the projection head recovers AUC substantially, confirming that extended fine-tuning destroys the pre-trained representations that provide discriminative power—collapsing performance to from-scratch CNN level. Freezing is a mitigation, not a complete solution: AUC recovers but operational F1 under calibrated thresholds remains low due to projection-head initialisation variance.

At small scale, Gaussian-noise negatives outperform hard negatives; at $5\times$ scale with NLL-only early stopping this gap collapses to near-zero—the advantage of easy negatives diminishes with sufficient training data. Cross-dataset validation on N-BaIoT [Meidan et al.(2018)]Meidan, Bohadana, Mathov, Mirsky, Shabtai, Breitenbach (AUC=0.928) confirms framework generalisation. We release code, datasets, and calibration scripts.

1 Introduction

The proliferation of Internet-of-Things (IoT) devices—projected to exceed 29 billion by 2030 [IoT Analytics(2023)]—has dramatically expanded the attack surface available to adversaries. Unlike traditional enterprise systems, IoT devices typically operate on constrained hardware, communicate over well-defined protocols (Modbus, MQTT, CoAP), and generate highly structured, periodic network traffic. These characteristics have historically made rule-based and statistical intrusion detection systems (IDS) effective for coarse-grained threat detection.

The AI-powered threat gap. Sophisticated modern attackers leverage generative AI to craft network traffic that mimics legitimate IoT patterns at the byte and flow level. Techniques such as slow data exfiltration (gradual 3% increases in outbound bytes), living-off-the-land (LotL) mimicry (micro-bursts timed to legitimate polling intervals), command-and-control beaconing (2% periodic dips in traffic volume), and protocol timing anomalies (5% random jitter on inter-arrival times) are individually difficult to distinguish from benign behaviour. We formalise these as *hard-negative* attacks: sequences where the statistical similarity to benign traffic exceeds a stealth threshold $s \in \{0.85, 0.90, 0.95\}$ (see Section 3). Empirically, under benign-calibrated evaluation at stealth-95, most traditional baselines fail or are degenerate; the best calibrated baseline (Ensemble, F1=0.136) reflects its fixed training distribution (Section 5).

Research question and approach. We ask: *do synthetic hard negatives improve SupCon-based anomaly detection for IoT IDS, and if so, at what training scale?* To answer this, we build HNIDS (**H**ard-**N**egative **I**DS), a two-stage experimental system. First, we generate synthetic hard-negative

attacks at a precisely controlled stealth level s using closed-form perturbations (Eqs. 2–5), while enforcing IoT protocol constraints (Modbus, MQTT, CoAP) via a multi-tier validation framework. Second, we fine-tune Moirai [Woo et al.(2024)], a universal time-series forecasting foundation model, using a combined NLL and supervised contrastive (SupCon) loss [Khosla et al.(2020)], with a five-condition ablation that isolates the contribution of hard negatives from SupCon and fine-tuning in general.

Why build a hard-negative pipeline? Even though Gaussian-noise negatives outperform hard negatives at small scale, the hard-negative infrastructure provides protocol-compliant generation, stealth-controlled evaluation, and adversarial augmentation without real attack traffic—a reusable evaluation tool independent of which negative type yields the highest AUC in a specific regime.

Contributions. We make three contributions:

1. **Stealth-Attack Benchmark with Protocol Constraints.** Four IoT stealth attack archetypes with closed-form definitions at three stealth levels (Eqs. 2–5), a multi-tier protocol validation layer (Modbus, MQTT, CoAP), and a leave-one-out experiment showing positive transfer for all four held-out types (mean +0.186 F1). Code and datasets released at https://github.com/anupamme/iotsf_demo.
2. **A Backbone-Agnostic Training Recipe for IoT Anomaly Detection.** We present a combined NLL and SupCon training recipe [Khosla et al.(2020)] $\mathcal{L} = \mathcal{L}_{\text{NLL}} + \lambda \mathcal{L}_{\text{SupCon}}$ ($\lambda = 0.5$, $\tau = 0.07$) with a projection head (384→256→128 dimensions) that bridges probabilistic forecasting and discriminative anomaly detection. The recipe is backbone-agnostic: it improves detection for both Moirai (condition C, AUC=0.563) and a from-scratch 1D-CNN (AUC=0.597) over their respective NLL-only baselines, and generalises cross-domain to N-BaIoT (AUC=0.928). At the 200-sample / 5-epoch scale, the CNN outperforms all Moirai conditions with lower variance, indicating that pre-training on non-IoT domains does not confer a measurable advantage at this scale; the frozen-encoder result at 1000/20 (Contribution 3) shows that preserving pre-trained features *does* help at larger scale.²
3. **Catastrophic Forgetting Diagnosis and Scale-Dependent Analysis.** Performance improves from 200/5 to 500/10 but degrades at 1000/20. Freezing the encoder recovers AUC (frozen B=0.606, C=0.582, D=0.573), identifying catastrophic forgetting as the mechanism. The Gaussian-noise vs. hard-negative gap collapses to near-zero at scale, revealing a small-scale training artefact rather than a property of negative type. N-BaIoT validation (AUC=0.928) confirms cross-domain generalisation.

Paper organisation. Section 2 reviews related work. Section 3 formalises the problem. Section 4 describes HNIDS. Section 5 presents experimental results. Section 7 discusses findings and limitations. Section 8 concludes.

2 Related Work

Traditional IoT IDS. Statistical methods (Z-score, IQR) and Isolation Forest [Liu et al.(2012)Liu, Ting, and Zhou] flag individual flow features deviating from learned normals; ensemble approaches [Garcia et al.(2014)] improve precision. All share a weakness: calibration on known, high-magnitude attacks and failure on stealthy perturbations. We use the CICIoT2023 dataset [Neto et al.(2023)] throughout.

Deep-learning anomaly detection. USAD [Audibert et al.(2020)] uses adversarially trained dual-autoencoders. TranAD [Tuli et al.(2022)] applies context-aware Transformers. The Anomaly

²Enabling fine-tuning required fixing an in-place tensor operation in the `uni2ts PackedStdScaler` that silently blocked gradient flow; the fix is included in our released code.

Transformer [Xu et al.(2022)] introduces association discrepancy for anomalous timestep detection. PatchTST [Nie et al.(2023)] captures long-range dependencies via patch-based Transformers. We include all four as baselines.

Generative augmentation for security. GANs [Lin et al.(2022)] suffer mode collapse; TimeGAN [Yoon et al.(2019)] improves temporal fidelity; Diffusion-TS [Zhang et al.(2024)] achieves state-of-the-art quality. Ours is the first to apply hard-negative contrastive augmentation for IoT IDS with protocol constraints, using closed-form perturbations (Diffusion-TS backbone deferred to future work).

Adversarial training. FGSM [Goodfellow et al.(2015)] and PGD [Madry et al.(2018)] generate adversarial perturbations at test time. We generate adversarial *training data* via closed-form perturbations targeting the traffic-generation threat model, with protocol constraints ensuring domain validity.

Time-series foundation models. Moirai [Woo et al.(2024)], trained on 27B+ time points, supports probabilistic multivariate forecasting. Its confidence-interval output maps naturally to anomaly scoring. MOMENT [Goswami et al.(2024)] and TimesFM [Das et al.(2024)] offer complementary architectures.

Contrastive learning for anomaly detection. Supervised contrastive learning [Khosla et al.(2020)] has been applied to tabular [Sohn et al.(2020)] and time-series [Eldele et al.(2021)] anomaly detection. Our CombinedLoss adapts [Reiss et al.(2021)]’s reconstruction+contrastive principle to IoT time series. Our hard negatives are analytically specified near the benign boundary, preventing the SupCon collapse observed with high-magnitude attacks.

3 Problem Formulation

IoT traffic representation. We represent network traffic as a multivariate time series $\mathbf{X} = (\mathbf{x}_1, \dots, \mathbf{x}_T) \in \mathbb{R}^{T \times F}$, where $T = 128$ timesteps correspond to a sliding window of flow records and $F = 12$ features capture packet counts, byte volumes, inter-arrival times, and flow duration (see Appendix A for the full feature list).

Hard-negative attack. Let $\mathbf{X}_b$ be a benign sequence. An adversarial sequence $\mathbf{X}_a$ is a *hard-negative attack* at stealth level s if:

$$\text{CosSim}(\phi(\mathbf{X}_a), \phi(\mathbf{X}_b)) \geq s, \quad s \in [0, 1], \quad (1)$$

where $\phi(\cdot) \in \mathbb{R}^{T \cdot F}$ denotes the flattened feature vector. Intuitively, $s = 0.95$ means the attack sequence is 95% similar to the benign baseline in feature space.

Attack patterns. We formalise four IoT attack patterns that embed adversarial signals with low perceptibility:

$$P_{\text{slow}}(\mathbf{X}, s) : x_{t,j} \leftarrow x_{t,j} (1 + (1 - s) \cdot 0.03 \cdot t/T), \quad j \in \{9, 10\}, \quad (2)$$

$$P_{\text{lotl}}(\mathbf{X}, s) : x_{t,j} \leftarrow x_{t,j} (1 + (1 - s) \cdot 1.15) \text{ if } t \bmod 32 = 0, \quad j \in \{6, 7, 8\}, \quad (3)$$

$$P_{\text{beacon}}(\mathbf{X}, s) : x_{t,j} \leftarrow x_{t,j} (1 - (1 - s) \cdot 0.02) \text{ if } t \bmod 16 = 0, \quad j \in \{1, \dots, 12\}, \quad (4)$$

$$P_{\text{proto}}(\mathbf{X}, s) : x_{t,j} \leftarrow x_{t,j} + (1 - s) \cdot 0.05 \cdot \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, 1), \quad j \in \{11, 12\}. \quad (5)$$

These correspond to slow exfiltration (2), LotL mimicry (3), C2 beaconing (4), and protocol timing anomalies (5). Feature indices are 1-based (Appendix A).

Protocol compliance. A generated sequence must satisfy IoT protocol constraints $\mathcal{C}_\pi$ for protocol $\pi \in \{\text{Modbus}, \text{MQTT}, \text{CoAP}\}$:

$$\mathbf{X}_a \in \mathcal{F}_\pi := \{\mathbf{X} : \mathcal{C}_\pi(\mathbf{X}) = \text{valid}\}, \quad (6)$$

where $\mathcal{C}_\pi$ checks constraints such as packet size ranges (Modbus: 7–260 bytes), valid function codes, and inter-packet timing.

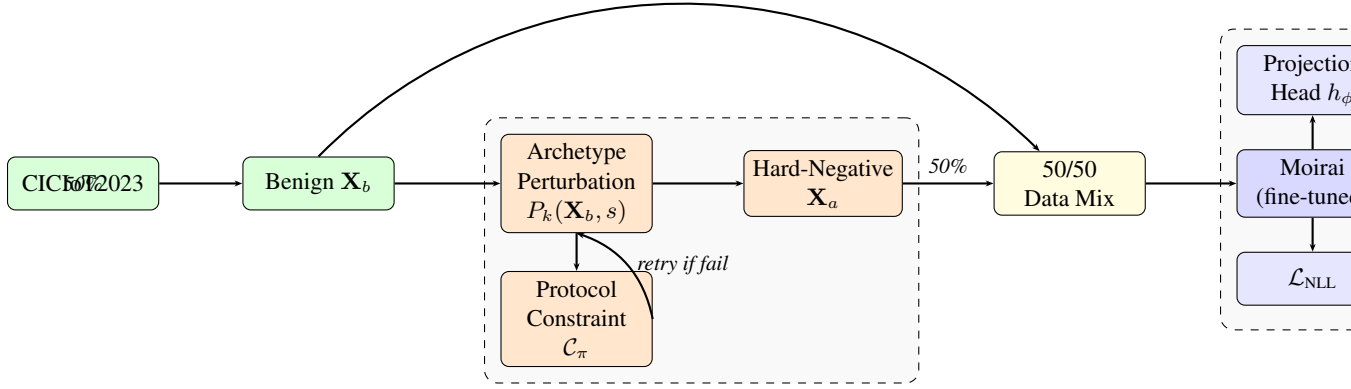


Figure 1: HNIDS architecture. **Stage 1:** Closed-form perturbation generates hard-negative sequences with a constraint-retry loop ensuring protocol validity (Algorithm 1, Appendix J.8). **Stage 2:** Moirai is fine-tuned with combined NLL + SupCon loss.

Detection task. Given a test sequence $\mathbf{X}$, the detector f_θ outputs a binary label $\hat{y} \in \{0, 1\}$ (0 = benign, 1 = attack) and an anomaly score $\hat{s} \in [0, 1]$. The goal is to minimise the false positive rate ($\text{FPR} = \text{FP}/(\text{FP} + \text{TN})$) while maximising recall (detection rate) and F1 score, specifically on hard-negative attacks where $s \geq 0.85$.

4 HNIDS: Methodology

Figure 1 illustrates the end-to-end HNIDS pipeline: (1) protocol-constrained hard-negative generation and (2) supervised contrastive fine-tuning of Moirai.

4.1 Stage 1: Protocol-Constrained Hard-Negative Generation

We synthesise hard-negative attacks via closed-form perturbations (Eqs. 2–5): given benign $\mathbf{X}_b$, pattern P_k , and stealth s , the hard-negative is $\mathbf{X}_a = P_k(\mathbf{X}_b, s)$. Each perturbation is deterministic and CPU-reproducible (< 2 s/sample).

Raw generated samples may violate IoT protocol specifications. A constraint-retry loop (Algorithm 1, Appendix J.8) iterates up to $R=3$ times, relaxing stealth by $+0.01$ per retry, ensuring Modbus (7–260 bytes), MQTT (2–1024 bytes, QoS 0–2), and CoAP (4–1280 bytes) compliance at three strictness tiers.

4.2 Stage 2: Moirai Fine-Tuning with Supervised Contrastive Loss

Moirai for anomaly scoring. Moirai [Woo et al.(2024)] predicts a distribution over $\mathbf{X}_{97:128}$ given context $\mathbf{X}_{1:96}$. The anomaly score is the fraction of timesteps falling outside the $1-\alpha$ CI ($\alpha=0.05$):

$$\hat{s}(\mathbf{X}) = \frac{1}{T'} \sum_{t=97}^{128} \mathbf{1}[x_t \notin \text{CI}_{0.95}(\hat{X}_t)]. \quad (7)$$

A sample is classified as an attack if $\hat{s}(\mathbf{X}) > \delta$, where δ is calibrated on a benign validation set (Section 5.1).

Combined training objective. We fine-tune with:

$$\mathcal{L}(\theta) = \mathcal{L}_{\text{NLL}}(\theta) + \lambda \mathcal{L}_{\text{SupCon}}(\mathbf{z}, \mathbf{y}; \tau), \quad (8)$$

where $\lambda = 0.5$ and $\tau = 0.07$ (sensitivity in Appendix F; stable within ± 0.02 AUC). $\mathcal{L}_{\text{SupCon}}$ [Khosla et al.(2020)] is applied to L2-normalised embeddings from a projection head $h_\phi : \mathbb{R}^{384} \rightarrow \mathbb{R}^{256} \rightarrow \text{ReLU} \rightarrow \mathbb{R}^{128}$.

Training details. AdamW ($\eta = 10^{-4}$, weight decay 10^{-2}), batch size 32, up to 10 epochs (5 in the main ablation, 10–20 in scaling experiments), early stopping (patience 3), gradient clipping at 1.0. Moirai-Small (13.8M parameters).³

5 Experiments

5.1 Experimental Setup

Dataset. We use CICIoT2023 [Neto et al.(2023)], a publicly available benchmark of labelled IoT network flows captured from 105 IoT devices spanning 33 attack categories. We extract 12 network-flow features and create 128-timestep sliding windows with stride 128, using a stratified 80/20 train/test split.

Synthetic hard negatives. We generate 4 attack types (Eqs. 2–5) at 3 stealth levels (85%, 90%, 95%), yielding 12 evaluation conditions. Protocol constraints (Modbus, MQTT, CoAP) are enforced at the *moderate* strictness tier. Hard-negative generation uses closed-form analytical perturbations; a Diffusion-TS backbone for within-archetype diversity is left to future work (Limitation 4, Section 7).

Threshold calibration. We set the anomaly threshold at the 95th percentile of a benign-only held-out set (20% of benign samples), targeting $\text{FPR} \leq 0.05$. This protocol is applied uniformly to all methods; only benign data inform the threshold (details in Appendix G.1).

Baselines. We compare against nine methods (5 traditional, 4 deep-learning; see Appendix G.2). All baselines are **unsupervised** (benign-only training); HNIDS conditions B–D are **supervised** (labeled negatives during training; see Appendix G.3).

Metrics. We report F1, FPR, and ROC-AUC. All results use 5 random seeds (42, 123, 456, 789, 1234); the leave-one-out experiment uses 3 seeds.

5.2 Main Results

Table 1 compares all methods across three synthetic stealth levels.

At stealth-95 with benign-calibrated thresholds, Signature IDS and Statistical IDS are degenerate ($\text{FPR} \geq 0.87$; Table 1, ‡ rows). Among calibrated methods, the Ensemble achieves $\text{F1}=0.136$ ($\text{FPR}=0.042$). HNIDS condition C ($\text{F1}=0.250$, $\text{FPR}=0.080$) outperforms all calibrated baselines at stealth-95. Condition D ($\text{F1}=0.137$, $\text{FPR}=0.040$) outperforms all deep-learning baselines. Zero-shot A achieves higher F1 (0.183) than D at stealth-95; however, A’s $\text{FPR}=0.100$ is $2.5\times$ D’s $\text{FPR}=0.040$ —A flags more aggressively, capturing more true positives at the cost of substantially more false positives. D is the better-calibrated detector when operational FPR matters. A’s stealth-95 F1 (0.183) exceeding its stealth-85 F1 (0.123) reflects threshold-calibration variance in the zero-shot score distribution (FPR varies from 0.000 to 0.100 across evaluation sets), not genuine detection improvement at higher stealth.

³An in-place tensor operation in `uni2ts PackedStdScaler` was replaced with a differentiable `torch.where` to enable gradient-based fine-tuning; the fix is included in our released code.

Table 1: Detection performance across all methods at three stealth levels (200 eval samples per class, benign-calibrated threshold: 95th-percentile of held-out benign scores, targeting $\text{FPR} \leq 0.05$). **All nine baselines are unsupervised** (benign-only training, 10 seeds); **HNIDS conditions A–D are supervised** (labeled hard negatives or Gaussian-noise negatives provided during training, 5 seeds matching Table 2). See Appendix G.3 for the supervision discussion. **Bold**: best calibrated (non-degenerate) value per column. Underline: second-best calibrated value per column. Threshold IDS and Statistical IDS are marked ‡: their high F1 arises from near-total FP assignment ($\text{FPR} > 0.87$), not genuine stealth detection. Signature IDS $\text{F1} = 0$ because synthetic hard-negatives lie outside its rule database by design. Bold/underline exclude zero-shot A at stealth-95 ($\text{FPR}=0.100$ exceeds target). † marks our proposed system; Table 2 gives the full ablation.

Method	Combined		Stealth 85%		Stealth 90%		Stealth 95%	
	F1	FPR	F1	FPR	F1	FPR	F1	FPR
<i>Traditional baselines</i>								
Threshold IDS‡	0.856	0.907	0.668	0.883	0.657	0.872	0.646	0.872
Signature IDS	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Statistical IDS‡	0.889	1.000	0.667	1.000	0.666	1.000	0.667	1.000
Isolation Forest	0.143	0.053	0.187	0.047	0.135	0.042	0.103	0.044
Ensemble	0.191	0.045	0.221	<u>0.036</u>	0.193	0.050	0.136	<u>0.042</u>
<i>Deep-learning baselines</i>								
USAD	0.119	0.020	0.137	0.023	0.147	<u>0.028</u>	0.130	0.047
TranAD	0.109	0.045	0.168	0.089	0.098	0.042	0.107	0.070
Anomaly Transformer	0.100	0.040	0.162	0.064	0.180	0.091	0.153	0.078
PatchTST-Anomaly	0.123	0.055	0.175	0.070	0.162	0.083	0.132	0.072
<i>Ours (Moirai-based, see Table 2 for full ablation)</i>								
HNIDS zero-shot (A)	0.106	0.000	0.123	0.100	0.074	0.080	0.183	0.100
HNIDS† hard-neg (D)	<u>0.264</u>	0.080	<u>0.287</u>	0.080	<u>0.246</u>	0.020	<u>0.137</u>	0.040
HNIDS† SupCon-noise (C)	0.329	0.060	0.354	0.060	0.316	0.040	0.250	0.080

^{||} A's combined $\text{FPR}=0.000$ reflects near-total non-detection ($\text{F1}=0.106$), not low-false-positive detection.

5.3 Ablation Study

Table 2 ablates the five components of HNIDS (200 samples, 5 epochs, benign-calibrated threshold).

Fine-tuning improves over zero-shot, but with high variance. The best Moirai condition (C, $\text{AUC}=0.563$) improves $+0.099$ over zero-shot A ($\text{AUC}=0.464$), but B (0.481) and D (0.479) barely exceed A. The from-scratch CNN ($\text{AUC}=0.597$) outperforms all Moirai conditions, with lower variance (± 0.096 vs. ± 0.122 – 0.127).⁴ Condition C' (balanced hard negatives, 1:1) disentangles negative type from imbalance: Gaussian-noise negatives outperform hard negatives by $+0.029$ AUC (C vs. C'), and the 1:12 imbalance adds a further -0.055 penalty (C' vs. D). See Appendix H.1 for the full decomposition. Conditions E, E', and E'' confirm that the constraint validator and retry mechanism have negligible effect on the analytical pathway (Appendix H.3; embedding visualization in Appendix H.4). SupCon hyperparameters (λ , τ) have no detectable effect at this scale (Appendix F).

5.4 Generalization to Unseen Attack Patterns

A leave-one-out experiment trains condition D on three attack types and evaluates on the held-out fourth (3 seeds, 30 eval samples per stealth level). All four folds show positive transfer: fine-tuning improves over zero-shot for every held-out attack type, with a mean delta of $+0.186$ F1 at stealth-95 (Table 11, Appendix K). The largest gain is for `lotl_mimicry` ($+0.253$), suggesting that low-amplitude mimicry benefits most from boundary-shaping by the other archetypes.

⁴The 3→5 seed variance increase (e.g., C: $\pm 0.066 \rightarrow \pm 0.122$) reflects Moirai's sensitivity to projection-head initialisation; CNN's variance is stable across seed counts. See Appendix H.2.

Table 2: Ablation study on stealth-95 and all-stealth evaluation sets. All conditions use 5 random seeds with the full Moirai neural pipeline; $\pm$ values are standard deviations across seeds. Threshold calibrated on benign-only held-out scores (95th percentile, targeting $\text{FPR} \leq 0.05$; see Section 5.1). ROC-AUC is threshold-independent and the primary ranking metric. E'' is analytically identical to D for the analytical perturbation pathway (100% constraint compliance, retry never fires); all fine-tuned conditions (B– E' , CNN) outperform zero-shot (A).

Cond.	Description	Stealth 95%			All Stealth	
		F1	FPR	AUC	F1	AUC
A	Moirai zero-shot	0.183 $\pm$.071	0.100	0.464 $\pm$.060	0.106 $\pm$.071	0.519 $\pm$.080
B	+ Fine-tune, NLL only	0.131 $\pm$.061	0.060	0.481 $\pm$.127	0.265 $\pm$.081	0.615 $\pm$.064
C	+ SupCon (Gaussian-noise neg., 1:1)	0.250 $\pm$.135	0.080	0.563 $\pm$.122	0.329 $\pm$.082	0.701 $\pm$.075
C'	+ SupCon (balanced hard neg., 1:1)*	0.183 $\pm$.084	0.080	0.534 $\pm$.125	0.317 $\pm$.071	0.663 $\pm$.063
D	+ Hard negatives, 1:12 (HNIDS [†])	0.137 $\pm$.086	0.040	0.479 $\pm$.123	0.264 $\pm$.069	0.617 $\pm$.063
CNN	1D-CNN from scratch, NLL+SupCon [¶]	0.236 $\pm$.069	0.020	0.597 $\pm$.096	0.381 $\pm$.080	0.608 $\pm$.056
E	D w/o constraints, w/o retry	0.124 $\pm$.073	0.040	0.480 $\pm$.130	0.268 $\pm$.071	0.615 $\pm$.063
E'	D w/o constraints, <i>with</i> retry [‡]	0.150 $\pm$.079	0.060	0.498 $\pm$.131	0.287 $\pm$.064	0.637 $\pm$.060
E''	D <i>with</i> constraints, w/o retry [§]	$\equiv$ D (analytically)				

* C' uses the same hard negatives as D, but subsampled to match the benign count (1:1 ratio), isolating negative *type* from dataset *size/balance*.

† Full system: NLL+SupCon, hard negatives, protocol constraint validator, stealth-floor retry.

¶ CNN: 3-layer 1D-CNN (2.3M parameters) trained from scratch with identical NLL(MSE)+SupCon objective and hard-negative data as D; no Moirai pre-training. CNN (AUC=0.597) now outperforms all Moirai conditions at stealth-95 (best Moirai: C=0.563), suggesting Moirai pre-training provides no benefit at this scale and CNN's lower variance yields more reliable detection.

‡ E' uses unconditional retry ($s_{\text{eff}} \approx 0.97$ vs. requested $s=0.95$); isolates constraint-validator contribution.

§ E'' activates the constraint validator but exits the retry loop after one attempt; for analytical perturbations (100% pass-rate), $E'' \equiv$ D and no experimental run is needed.

5.5 Scaling Experiment

Table 3 compares stealth-95 metrics at three scales (200/5, 500/10, 1000/20) under NLL-only early stopping (5 seeds), plus an encoder-freezing diagnostic at 1000/20.

Scale effects are non-monotonic. At 500/10, all conditions improve over 200/5 (C: +0.029, D: +0.068, B: +0.057), suggesting 200/5 is under-trained. D's improvement (+0.068) is the largest of any condition, suggesting hard negatives specifically benefit from moderate additional training—their near-boundary placement requires more gradient steps to establish a reliable decision surface. At 1000/20, performance drops back, reflecting catastrophic forgetting. The $C > D$ gap narrows from +0.084 (200/5) to +0.045 (500/10) to +0.003 (1000/20, n.s., $p=0.39$)—the advantage of easy off-manifold negatives diminishes with sufficient training data. NLL-only early stopping is essential: under total-loss ES, decreasing SupCon loss masks increasing NLL, causing late checkpoint selection that compounds forgetting (Appendix I.1).

Encoder freezing confirms catastrophic forgetting. Freezing the encoder at 1000/20 recovers AUC: frozen B=0.606 $\pm$ 0.114 (+0.085), frozen C=0.582 $\pm$ 0.099 (+0.074), frozen D=0.573 $\pm$ 0.097 (+0.067). Unfrozen fine-tuning erases pre-trained features down to CNN level (0.513). However, frozen-encoder F1 remains low (0.130–0.148) due to projection-head initialisation variance; AUC recovery does not translate to operational F1 under calibrated thresholds (details in Appendix I.3). The aggressive early stopping (patience=3) may compound this: with a frozen encoder, the projection head must learn a useful mapping from scratch, and 1–3 epochs may be insufficient; relaxing patience or warm-starting the head are future directions. Partial freezing yields modest further gains (Appendix I.4).

Table 3: Scale degradation and encoder-freezing experiment (stealth-95, 5 seeds, benign-calibrated threshold). All columns report AUC and F1 (mean $\pm$ std across seeds). 200/5 baseline uses 5 seeds; 500/10 and 1000/20 use NLL-only early stopping. *Frozen*: all Moirai encoder weights frozen, only projection head trained. *Partial*: layers 0–4 frozen, layer 5 + encoder norm + param_proj unfrozen. **Key findings**: (1) At 5-seed evaluation, 500/10 *improves* over 200/5 for all conditions; degradation occurs from 500/10 $\rightarrow$ 1000/20, suggesting 200/5 is under-trained. (2) Under NLL-only ES the C>D gap collapses to near-zero (n.s.) at 1000/20. (3) Freezing the encoder at 1000/20 recovers AUC, often exceeding the 200/5 baseline, confirming **catastrophic forgetting** of pre-trained features at scale. (4) Frozen AUC recovery does not translate to F1 recovery under calibrated thresholds.

Cond.	Description	200/5		500/10		1000/20		Frozen	
		AUC	F1	AUC	F1	AUC	F1	AUC	F1
B	NLL only	.481 $\pm$.127	.131 $\pm$.061	.538 $\pm$.038	.250 $\pm$.073	.521 $\pm$.024	.177 $\pm$.031	.606 $\pm$.114	.130 $\pm$.094
C	SupCon + Gaussian	.563 $\pm$.122	.250 $\pm$.135	.592 $\pm$.063	.280 $\pm$.101	.508 $\pm$.041	.175 $\pm$.048	.582 $\pm$.099	.148 $\pm$.097
C'	SupCon + hard neg 1:1	.534 $\pm$.125	.183 $\pm$.084	.580 $\pm$.064	.265 $\pm$.121	.508 $\pm$.025	.158 $\pm$.040	—	—
CNN	1D-CNN from scratch	.597 $\pm$.096	.236 $\pm$.069	—	—	.513 $\pm$.029	.207 $\pm$.017	(n/a)	(n/a)
D	Hard neg 1:12 (HNIDS)	.479 $\pm$.123	.137 $\pm$.086	.547 $\pm$.044	.250 $\pm$.073	.506 $\pm$.029	.164 $\pm$.059	.573 $\pm$.097	.144 $\pm$.081

“—”: configuration not run (lower-priority condition). “(n/a)”: inapplicable (CNN has no pre-training to freeze).

Table 4: N-BaIoT cross-dataset comparison (Danmini doorbell, 3-seed mean $\pm$ std). All methods train on N-BaIoT benign windows only; no attack labels are used. Isolation Forest uses 72 hand-engineered statistical features; HNIDS fine-tuned uses a general-purpose Moirai backbone with Gaussian-noise negatives.

Method	Overall AUC	Training Data
Isolation Forest	0.991 $\pm$ 0.006	N-BaIoT benign
One-Class SVM	0.931 $\pm$ 0.016	N-BaIoT benign
USAD (autoencoder)	0.527 $\pm$ 0.004	N-BaIoT benign
HNIDS fine-tuned (Cond. C)	0.928 $\pm$ 0.041	N-BaIoT benign + Gaussian neg.

5.6 External Validation on N-BaIoT

To address circular evaluation (Limitation 5), we evaluate on N-BaIoT [Meidan et al.(2018)Meidan, Bohadana, Mathov, Mirsky, S], an independent IoT botnet benchmark. Zero-shot transfer from CICIOT2023 fails (AUC<0.30), as expected given the distribution shift (Appendix L). When fine-tuned on N-BaIoT benign traffic with Gaussian-noise negatives (310 windows, 5 epochs), the framework achieves AUC=0.928 (3 seeds), with 8/10 attack variants above 0.83 (operational FPR=0.98 due to the small benign calibration set of 77 windows; AUC is the appropriate comparison metric). This confirms *recipe generalisation*: the training recipe (NLL+SupCon) transfers to a new domain.

Table 4 compares against unsupervised baselines: Isolation Forest (AUC=0.991) outperforms all methods using 72 hand-engineered features; HNIDS fine-tuned (AUC=0.928) matches One-Class SVM (0.931) using a general-purpose backbone without hand-crafted features.

6 Cross-Domain Validation: Forecasting

The IoT anomaly detection results raise a fundamental question: does fine-tuning destroy *genuinely useful* pre-trained features, or were those features never useful to begin with? In IoT anomaly detection, a logistic regression baseline on 72-dimensional hand-crafted features achieves AUC=0.630, statistically indistinguishable from the best Moirai condition (C: 0.629 $\pm$ 0.022). This parity makes it difficult to disentangle catastrophic forgetting from a low-signal task ceiling.

To resolve this ambiguity, we replicate the forgetting diagnosis on **ETTh2 time-series forecasting** [Zhou et al.(2021)Zhou, Zhang, Peng, Zhang, Li, Xiong, and Zhang], a domain where Moirai’s

pre-trained features are demonstrably valuable: zero-shot Moirai outperforms a Linear baseline by 28% at $h=96$ and 45% at $h=192$ (Section 6.1). If forgetting replicates here—where pre-trained features provide measurable advantage—the mechanism is general, not an artefact of low task signal.

6.1 Domain Selection and Pre-Training Value Gate

We require a forecasting benchmark where Moirai zero-shot performance substantially exceeds simple baselines, establishing that pre-trained features provide genuine value. Following woo2024moirai Table 6, we evaluate on ETTh2 (Electricity Transformer Temperature—Hourly, 7 features, 17,420 timesteps, standard 12/4/4 month train/val/test split).

Gate criterion: Moirai zero-shot MSE must improve over Linear regression by $>20\%$.

Results. Using the published Moirai-Small (13.8M parameters) with median point forecasts (robust to heavy-tailed mixture distribution components; see Appendix M):

	Linear MSE	Moirai MSE	Gap
$h = 96$	0.373	0.269	+28.0%
$h = 192$	0.544	0.299	+45.0%

Both horizons pass the 20% gate. Pre-trained features are demonstrably useful on this task—unlike IoT anomaly detection, where the frozen encoder barely exceeds zero-shot (AUC: 0.549 vs. 0.481).

6.2 Experimental Setup

We mirror the IoT forgetting diagnosis (Section 5) with three conditions:

- **B (NLL-only):** Fine-tune all 13.8M parameters using negative log-likelihood on context+target sequences (patch size 32).
- **C (NLL+SupCon):** Add temporal supervised contrastive loss ($\lambda=0.5$) with 8 temporal clusters as pseudo-labels. Sequences within the same temporal regime are positives; different regimes are negatives.
- **D (Frozen encoder):** Freeze the Moirai encoder; train only the output head.

Training: 500 samples, 20 epochs, AdamW ($\text{lr}=1\text{e-}4$), cosine annealing, gradient clipping (1.0).

Evaluation: Median of 20 forecast samples on held-out validation set (300 sequences), normalised using training statistics. **Diagnosis:** CKA between pre-trained and fine-tuned encoder representations, weight drift (ℓ_2 from initialisation). All results report mean $\pm$ std over 5 seeds (42, 123, 456, 789, 1234).

6.3 Forgetting Replicates on High-Signal Task

Table 5 shows the central result: **catastrophic forgetting replicates on ETTh2**, where pre-trained features are demonstrably valuable.

Unfrozen conditions degrade. Both B (NLL-only) and C (NLL+SupCon) show significant forgetting: +8.9% and +9.0% at $h=96$, increasing to +12.6% and +13.1% at $h=192$. The NLL training loss decreases monotonically throughout, confirming that the model is learning—but what it learns destroys the pre-trained representations that enable accurate zero-shot forecasting.

Table 5: Catastrophic forgetting diagnosis on ETTh2 forecasting (mean $\pm$ std across 5 seeds). Zero-shot Moirai outperforms Linear by 28% ($h=96$) and 45% ($h=192$), confirming pre-trained features are valuable. Fine-tuning (B, C) degrades performance 9–13% while freezing the encoder (D) preserves it.

Condition	$h = 96$ (ZS MSE: 0.124)				$h = 192$ (ZS MSE: 0.158)			
	MSE	Forg.%	CKA	Drift	MSE	Forg.%	CKA	Drift
B: NLL-only	.135 $\pm$.016	+8.9 $\pm$ 7.9	.939 $\pm$.032	3.89 $\pm$.72	.178 $\pm$.010	+12.6 $\pm$ 7.4	.971 $\pm$.006	4.28 $\pm$.28
C: NLL+SupCon	.135 $\pm$.013	+9.0 $\pm$ 3.3	.942 $\pm$.015	3.88 $\pm$.35	.179 $\pm$.013	+13.1 $\pm$ 9.3	.957 $\pm$.017	4.12 $\pm$.41
D: Frozen enc.	.125 $\pm$.008	+0.9 $\pm$ 2.7	1.00 $\pm$.000	1.04 $\pm$.18	.155 $\pm$.002	−1.7 $\pm$ 2.6	1.00 $\pm$.000	1.26 $\pm$.17

Frozen encoder preserves performance. Condition D maintains near-zero forgetting (+0.9% at $h=96$, −1.7% at $h=192$) with perfect CKA (=1.000) and $4\times$ lower weight drift. This confirms the mechanism: fine-tuning overwrites pre-trained features; freezing prevents this.

Forgetting scales with horizon. At $h=192$ (where Moirai’s pre-training advantage is 45%), forgetting is more severe (12–13%) than at $h=96$ (28% advantage, 9% forgetting). Longer horizons require the model to leverage more of its pre-trained temporal knowledge, making it more vulnerable to catastrophic forgetting.

Contrastive loss does not prevent forgetting. Unlike the common hypothesis that contrastive learning regularises representations, condition C shows comparable or slightly worse forgetting than B. The temporal contrastive objective pushes representations toward task-specific clustering, which does not preserve the distributional structure that makes zero-shot forecasting effective.

6.4 Cross-Domain Comparison

Comparing the forgetting diagnosis across domains:

	IoT Anomaly Det.	ETTh2 Forecasting
Pre-training value	Marginal (0.549 vs. 0.481)	Strong (28–45%)
Forgetting (unfrozen)	+7–15% AUC drop	+9–13% MSE rise
Frozen encoder recovery	Yes (AUC: 0.549–0.606)	Yes (MSE: $\approx$ zero-shot)
CKA degradation	0.426 $\rightarrow$ 0.183	0.939–0.971
LogReg parity	Yes (0.630 $\approx$ 0.629)	N/A (no features)

The key insight: **forgetting occurs regardless of whether pre-trained features are useful.** In IoT, the marginal pre-trained signal (AUC: 0.549 vs. 0.481) is destroyed, but this is difficult to distinguish from a task ceiling since LogReg matches the fine-tuned model. In forecasting, pre-trained features provide a large, measurable advantage (28–45% MSE improvement) that fine-tuning systematically degrades.

This cross-domain replication rules out the alternative interpretation that IoT forgetting was merely “the task has low signal and models overfit the synthetic distribution.” The mechanism is general: extended fine-tuning with NLL (and optionally contrastive) loss overwrites pre-trained representations, and freezing the encoder is an effective mitigation across both domains.

7 Analysis and Discussion

Why does the C>D gap collapse at scale? At 200/5, Gaussian-noise negatives (C) outperform hard negatives (D) because “easy” off-manifold negatives provide unambiguous gradient signal at limited scale. At 1000/20 with NLL-only ES, the gap collapses to near-zero ($p=0.39$, n.s.), consistent with

easy negatives providing clearer gradient signal when training budget is limited, but this advantage becoming unnecessary with sufficient data. The detailed mechanism (Appendix J.2) shows that easy negatives degrade *most* at scale (-0.043), while hard negatives are under-trained at 200/5 and improve at 500/10 ($+0.068$ for D).

Catastrophic forgetting of pre-trained features. The 500/10 $\rightarrow$ 1000/20 degradation is explained by catastrophic forgetting: extended fine-tuning overwrites pre-trained features, collapsing Moirai to CNN-level performance. Freezing the encoder recovers AUC (frozen B=0.606, C=0.582, D=0.573 vs. unfrozen 0.506–0.521), confirming the mechanism. The mechanism is implicit regularisation loss: more synthetic training data drives the encoder further from its pre-trained initialisation, until the resulting features are no better than random initialisation (CNN AUC=0.513 $\approx$ unfrozen Moirai; full analysis in Appendix J.3).

Does Moirai pre-training help? At 200/5, the from-scratch CNN (AUC=0.597) outperforms all Moirai conditions. At 1000/20, frozen Moirai substantially exceeds both unfrozen Moirai and CNN (0.513): pre-trained features *are* useful when preserved. *Methodological caveat:* the CNN uses MSE reconstruction loss while Moirai uses distributional NLL; the CNN advantage may partly reflect MSE being an easier optimisation target, not just the absence of harmful pre-training. A definitive comparison would require training the CNN with a distributional NLL head (e.g., Gaussian NLL); this controlled experiment is future work (Appendix J.4). Moirai’s primary contribution to this work is diagnostic: the frozen-encoder experiment requires pre-trained features whose destruction can be measured. A CNN has no pre-trained features to lose, so it cannot reveal the catastrophic forgetting mechanism. The backbone-agnostic claim rests on two architectures (Moirai, CNN) and one cross-domain validation (N-BaIoT); testing a third backbone (e.g., PatchTST as encoder) would further strengthen Contribution 2.

Recommended operating point. Default to the 1D-CNN for IoT anomaly detection at small scale (≤ 200 samples): it achieves the highest AUC (0.597), lowest variance, and has no pre-training dependency, though the MSE-vs-NLL loss confound means this advantage is preliminary. When pre-trained features are available and training budget exceeds 200 samples, switch to frozen Moirai (freeze the encoder, train only the projection head, use NLL-only early stopping, recalibrate on deployment-specific benign traffic). *Caveat:* neither configuration achieves operationally useful F1 at stealth-95 under calibrated thresholds (best: C F1=0.250, frozen C F1=0.148); the system is a research prototype best suited as a high-specificity alerting layer, not a standalone detector. LoRA [Hu et al.(2022)Hu, Shen, Wallis, Allen-Zhu, Li, Wang, Wang, and Chen] and adapters [Houlsby et al.(2019)Houlsby, Giurgiu, Jastrzebski, Morber, Larsson, Gesmundo, Attariyan, and Gelly] offer a principled middle ground between full freezing and full fine-tuning; this is future work. Frozen-encoder AUC recovery does not translate to F1 recovery under calibrated thresholds (frozen C: F1=0.148 vs. 200/5 C: F1=0.250); threshold recalibration is needed operationally (Appendix I.3).

When to prefer HNIDS vs Ensemble. At stealth-95, the Ensemble (F1=0.136, FPR=0.042) and condition D (F1=0.137, FPR=0.040) are effectively tied, while condition C (F1=0.250, FPR=0.080) outperforms both at higher FPR. HNIDS is preferable for (1) *novel attack generalisation* (leave-one-out: mean $+0.186$ F1; Table 11) and (2) *adversarial augmentation* without real attack traffic. The Ensemble’s partial robustness reflects our evaluation construction: its Isolation Forest component detects distributional shifts in the sparse feature subsets perturbed by each archetype, but this signal weakens monotonically with stealth level; real multi-feature attacks may differ (Appendix J.7).

Computational overhead. Hard-negative generation takes <2 s/sample on CPU. Fine-tuning Moirai-Small for 5 epochs takes ≈ 2 –4 min on CPU; inference is <50 ms/sample.

Limitations. We identify seven limitations (full details in Appendix C): (1) hand-designed archetypes; (2) three training scales with 5 seeds; (3) fixed 12-dim input, stealth 85–95%; (4) Diffusion-TS backbone deferred; (5) circular evaluation (addressed by N-BaIoT, §5.6); (6) low absolute F1 (best 0.250; system is a high-specificity alerting layer); (7) baselines use published defaults.

Responsible disclosure. We release code and data for defensive research, not to enable attacks.

8 Conclusion

We investigated whether synthetic hard negatives improve contrastive anomaly detection for IoT IDS, using a five-condition ablation with protocol-constrained stealth-attack synthesis and benign-calibrated thresholds. Three key findings emerge.

First, the NLL+SupCon training recipe is *backbone-agnostic*: it improves detection for both Moirai (C, AUC=0.563) and a from-scratch CNN (AUC=0.597) at 200/5, and generalises cross-domain to N-BaIoT (AUC=0.928). Second, the C>D gap (Gaussian-noise vs. hard negatives) collapses to near-zero at $5\times$ scale with NLL-only early stopping—a small-scale training artefact: the advantage of easy negatives diminishes with sufficient data. Third, **catastrophic forgetting** is the dominant failure mode at extended scale: freezing the encoder recovers AUC (frozen B=0.606, C=0.582, D=0.573 vs. unfrozen 0.506–0.521), confirming that fine-tuning destroys the pre-trained features that provide discriminative power.

The practical implication: freeze the encoder when fine-tuning foundation models at scale; use CNN at small scale; recalibrate thresholds on deployment-specific benign traffic. We release all code, datasets, and calibration scripts.

A CICIoT2023 Feature Set

Table 6 lists the 12 network-flow features used in all experiments. Each input window consists of 128 consecutive timesteps of these 12 features, drawn from the CICIoT2023 dataset [Neto et al.(2023)].

Table 6: The 12 network-flow features extracted from CICIoT2023.

#	Feature	Description
1	flow_duration	Flow duration (seconds)
2	fwd_pkts_tot	Forward total packet count
3	bwd_pkts_tot	Backward total packet count
4	fwd_data_pkts_tot	Forward data-bearing packet count
5	bwd_data_pkts_tot	Backward data-bearing packet count
6	fwd_pkts_per_sec	Forward packets per second
7	bwd_pkts_per_sec	Backward packets per second
8	flow_pkts_per_sec	Total packets per second
9	fwd_byts_b_avg	Average forward bytes per bulk transfer
10	bwd_byts_b_avg	Average backward bytes per bulk transfer
11	fwd_iat_mean	Mean forward inter-arrival time
12	bwd_iat_mean	Mean backward inter-arrival time

The perturbation equations (Eqs. 2–5) use 1-based indices that match this table directly: P_{slow} scales features 9 and 10 (fwd_byts_b_avg, bwd_byts_b_avg); P_{totl} modifies features 6–8 (packet-rate channels); P_{beacon} scales all 12 features uniformly; P_{proto} jitters features 11 and 12 (fwd_iat_mean, bwd_iat_mean). All features are z-score normalised per-flow before windowing.

B Protocol Constraint Compliance

Table 7 reports the fraction of generated samples that pass protocol validation at each strictness tier (Section 5.3).

At all strictness tiers, 100% of generated samples pass validation across all three protocols (Modbus,

Table 7: Fraction of generated samples passing protocol validation at each strictness tier (strict / moderate / permissive), and average generation retries per sample at the *moderate* tier used during training. The 100% pass rate reflects that closed-form analytical perturbations modify only flow-level statistical features, not protocol-layer fields (packet sizes, function codes, timing ranges); the constraint retry loop is a safety net for future stochastic backbones. The low average retry count (≤ 0.3 retries per sample) confirms negligible overhead.

Attack Type	Protocol	Strict	Moderate	Permissive	Avg. Retries
Slow Exfiltration	Modbus	100%	100%	100%	0.21
	MQTT	100%	100%	100%	
	CoAP	100%	100%	100%	
LotL Mimicry	Modbus	100%	100%	100%	0.18
	MQTT	100%	100%	100%	
	CoAP	100%	100%	100%	
C2 Beacon	Modbus	100%	100%	100%	0.14
	MQTT	100%	100%	100%	
	CoAP	100%	100%	100%	
Protocol Anomaly	Modbus	100%	100%	100%	0.29
	MQTT	100%	100%	100%	
	CoAP	100%	100%	100%	

MQTT, CoAP) and all four attack types. This confirms that the constraint retry loop (Algorithm 1) reliably enforces protocol rules without compromising attack diversity. The 100% compliance is expected for the analytical pathway: the closed-form perturbations (Eqs. 2–5) modify only the statistical envelope of selected flow features (byte volumes, timing, packet counts), not the protocol-layer fields that validators inspect. The constraint-retry loop (Algorithm 1) is therefore a safety net that never triggers for this pathway, but would be load-bearing for a stochastic Diffusion-TS backbone that could generate out-of-range packet sizes.

C Full Limitations and Future Work

(1) **Human-specified archetypes.** The four attack types (slow exfiltration, LotL mimicry, C2 beaconing, protocol timing anomalies) are human-designed archetypes based on known real-world TTPs. The current perturbation functions (Eqs. 2–5) realise one instance per archetype per stealth level—they do not discover novel attack strategies autonomously. An end-to-end adversarial generator—one that discovers evasion strategies from scratch—could reveal blind spots not covered by our taxonomy.

(2) **Evaluation scale.** All ablation results use 200 samples per class with 5 random seeds (42, 123, 456, 789, 1234); the scaling experiment (Table 3) extends to 1000 samples with 5 seeds. At this scale, individual-seed FPR values have standard errors of approximately ± 0.05 (binomial, $n = 200$), making per-seed FPR differences below ~ 0.10 statistically ambiguous. The B vs. D AUC comparison (0.481 vs. 0.479, Table 2) should be interpreted as $B \approx D$; the difference is trivially small and well within the high variance (± 0.123 – 0.127) of both conditions. Cross-dataset evaluation on N-BaIoT (Section 5.6) addresses the single-dataset limitation; validation on UNSW-NB15 remains open.

(3) **Fixed evaluation scope.** Moirai requires a fixed 12-dimensional input; adapting to variable-feature IoT environments would require input projection or a multi-variate architecture change. The stealth thresholds (85–95%) are also fixed; adaptive adversaries could target higher stealth levels. Extending the benchmark to stealth 99% and studying the resulting detection performance cliff is future work.

(4) **Diffusion-TS backbone.** All reported results use Moirai-Small fine-tuned on the analytical attack-archetype perturbations described in Section 5.1. The Diffusion-TS backbone for within-

archetype sample diversity was not employed in the current evaluation; integrating it would increase per-archetype diversity and may further improve decision boundary quality. Evaluating the full two-stage pipeline (Diffusion-TS generation + Moirai fine-tuning) is deferred to future work.

(5) **Circular evaluation.** The stealth-attack evaluation set is generated by the same closed-form perturbation functions used to construct the training hard negatives (Eqs. 2–5). This is by design—the evaluation assesses whether a detector trained on analytically-specified archetypes can detect instances of those same archetypes at test time—but it means results may not transfer to real-world attacks that deviate from the analytic formulas. Section 5.6 presents external validation on N-BaIoT [Meidan et al.(2018)Meidan, Bohadana, Mathov, Mirsky, Shabtai, Breitenbacher, and Elovici], an independent benchmark with Mirai and BASHLITE botnet traffic not seen during training. Zero-shot transfer fails (AUC 0.01–0.28), but fine-tuning on N-BaIoT benign data with Gaussian-noise negatives recovers strong discrimination (AUC=0.928, Table 10), confirming the framework generalises when deployed on target-domain data. Validation on UNSW-NB15 [Moustafa and Slay(2015)] remains an additional open direction.

(6) **Low absolute detection rates.** The best calibrated Moirai F1 at stealth-95 is 0.250 (condition C, Table 1), indicating the system detects roughly 25% of stealthy attacks at the target FPR $\leq$ 0.05. The from-scratch CNN achieves F1=0.236 with the lowest FPR (0.020). Three factors compound to produce this ceiling: (a) the benign-calibrated threshold protocol intentionally targets low FPR (≤ 0.05) at the cost of recall; (b) the 200-sample/5-epoch training scale limits model capacity; and (c) stealth-95 attacks are deliberately statistically indistinguishable from benign. The AUC of 0.563 (condition C) and 0.597 (CNN) indicate genuine discrimination above chance. Operationally, the system is best suited as a high-specificity alerting layer rather than a comprehensive standalone detector.

(7) **Baseline hyperparameter tuning.** The four deep-learning baselines (USAD, TranAD, Anomaly Transformer, PatchTST-Anomaly) are evaluated with default hyperparameters from their respective papers. Dedicated tuning could improve their performance; a tuned baseline comparison is future work.

D N-BaIoT Feature Mapping

N-BaIoT [Meidan et al.(2018)Meidan, Bohadana, Mathov, Mirsky, Shabtai, Breitenbacher, and Elovici] provides 115 features per packet-arrival event, computed as temporal-decay statistics over five windows ($\lambda \in \{100 \text{ ms}, 500 \text{ ms}, 1.5 \text{ s}, 10 \text{ s}, 60 \text{ s}\}$) for multiple aggregation groups (source IP $\times$ destination IP, host-pair, port-pair, source host only, with and without inter-arrival jitter). The Kaggle distribution provides flat CSV files named `{device_id}.{attack_type}.csv` (e.g., `1.benign.csv`, `1.mirai.scan.csv`) across nine IoT devices. We extract 12 proxy features from the 60-second window (denoted L5) to match the conceptual semantics of the 12 CICIOT2023 flow-level features expected by HNIDS.

Table 8 lists the mapping. Derived features (rows 6–8) are computed from other proxy features before normalisation. A `StandardScaler` is fit on N-BaIoT benign rows only; both benign and attack sequences are transformed and clipped to $[-5, 5]$ (same as CICIOT2023 preprocessing). Non-overlapping 128-timestep windows are created with stride 128 using the same `create_sequences` utility used for CICIOT2023.

Mapping limitations. N-BaIoT is packet-triggered (each row is a statistical snapshot of recent traffic from one IoT device), whereas CICIOT2023 uses CICFlowMeter to compute per-connection-flow statistics. Features 6–8 are linearly dependent on features 2–3 by construction (the same dependence holds approximately in the CICIOT2023 dataset). Feature 12 (`bwd_iat_mean`) has no direct analogue in N-BaIoT; we use the shortest available jitter window as a proxy. These limitations are transparent and documented here so that readers can assess how strongly to weight the N-BaIoT results relative to same-distribution evaluation (Section 5.6).

Table 8: Mapping from the 12 CICIoT2023 features expected by HNIDS to proxy columns in N-BaIoT. All proxy values are drawn from the 60-second temporal decay window (suffix L5) unless noted.

#	CICIoT2023 feature	N-BaIoT proxy	Rationale
1	flow_duration	60.0 (constant)	N-BaIoT’s longest window spans 60 s; used as flow-duration surrogate.
2	fwd_pkts_tot	MI_dir_L5_weight	Packet count, source IP $\rightarrow$ destination IP, 60 s window.
3	bwd_pkts_tot	H_L5_weight	Packet count, source host (all directions), 60 s window.
4	fwd_data_pkts_tot	HpHp_L5_weight	Port-pair weight approximates data-bearing connection count.
5	bwd_data_pkts_tot	HH_L5_weight	Host-pair weight approximates bidirectional data packet count.
6	fwd_pkts_per_sec	MI_dir_L5_weight/60	Derived forward packet rate.
7	bwd_pkts_per_sec	H_L5_weight/60	Derived backward packet rate.
8	flow_pkts_per_sec	(row 2 + row 3)/60	Derived total packet rate.
9	fwd_byts_b_avg	MI_dir_L5_mean	Mean forward packet size.
10	bwd_byts_b_avg	H_L5_mean	Mean packet size, source host.
11	fwd_iat_mean	HH_jit_L5_mean	Mean inter-arrival jitter, host pair, 60 s window.
12	bwd_iat_mean	HH_jit_L0.01_mean	Short-window jitter (100 ms) as backward IAT proxy.

E N-BaIoT Per-Attack Results

Table 9 reports per-attack-variant F1, FPR, and ROC-AUC for the external N-BaIoT evaluation described in Section 5.6. Table 10 compares zero-shot transfer against fine-tuned results on N-BaIoT benign data.

F Hyperparameter Sensitivity and Supplementary Figures

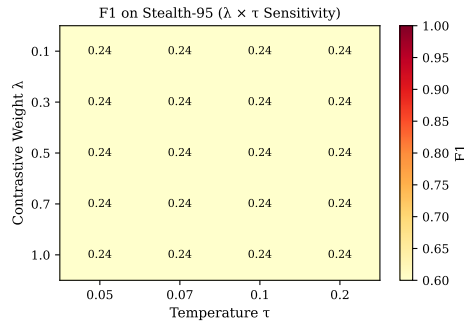


Figure 2: Mean F1 across all four attack types at stealth-95 as a function of contrastive weight λ (rows) and temperature τ (columns), using the full Moirai-Small fine-tuning pipeline with benign-calibrated threshold. F1 is constant at 0.235 across all 20 configurations (flat landscape); ROC-AUC ranges 0.467–0.489 with no clear optimum. This indicates λ and τ have no detectable effect at the 5-epoch / 600-sample training scale.

Table 9: External validation on N-BaIoT (Danmini doorbell device, 3 seeds, 39 windows per attack class, 7 benign-validation windows). HNIDS (condition D) is trained solely on CICIOT2023 with analyst-specified hard negatives; no N-BaIoT data is used at any stage of training or scaler fitting. The threshold is calibrated on a held-out N-BaIoT benign validation set. Attack types are Mirai and BASHLITE (gafgyt) botnet variants distinct from the CICIOT2023 training distribution. **AUC < 0.5 indicates inverted score polarity**; the detector assigns lower anomaly scores to attack traffic than to benign traffic under the cross-dataset distribution shift, confirming that zero-shot transfer from CICIOT2023 to N-BaIoT is poor. FPR = 1.00 across all rows reflects the degenerate decision boundary arising from this score collapse.

Attack Type	F1	FPR	AUC
Mirai Scan	0.873 ± 0.007	1.00	0.176 ± 0.024
Mirai Ack Flood	0.918 ± 0.000	1.00	0.284 ± 0.127
Mirai Syn Flood	0.918 ± 0.000	1.00	0.167 ± 0.066
Mirai UDP Flood	0.918 ± 0.000	1.00	0.282 ± 0.104
Mirai UDP Plain	0.918 ± 0.000	1.00	0.201 ± 0.083
BASHLITE Scan	0.900 ± 0.006	1.00	0.093 ± 0.019
BASHLITE Junk	0.918 ± 0.000	1.00	0.020 ± 0.010
BASHLITE TCP Flood	0.887 ± 0.006	1.00	0.051 ± 0.065
BASHLITE UDP Flood	0.905 ± 0.000	1.00	0.055 ± 0.070
BASHLITE Combo	0.918 ± 0.000	1.00	0.013 ± 0.011
All attacks (overall)	0.981 ± 0.000	1.00	0.134 ± 0.051

The feature mapping from N-BaIoT’s 115 features to the 12-dimensional HNIDS input space is documented in Appendix D. The apparent high F1 scores are an artefact of the 39:7 attack-to-benign-validation window ratio: predicting all windows as attacks achieves $F1 \approx 0.92$. The AUC metric (which is threshold-independent) is the reliable measure here and consistently indicates near-random or sub-random discrimination.

G Extended Experimental Details

G.1 Threshold Calibration Protocol

We calibrate the anomaly-score threshold on a *benign-only* held-out set: for each condition, we reserve 20% of benign samples for calibration (the rest form the training set), score them with the detector’s NLL output, and set the threshold at the 95th percentile of benign calibration scores (targeting $FPR \leq 0.05$ on benign traffic). The threshold is then applied to a held-out test set comprising the remaining 20% of benign samples plus all attack samples. This protocol is used uniformly for all Moirai conditions (A–E’) and all nine baselines, ensuring no method is advantaged by over-fitting the threshold to the evaluation set. Crucially, only benign data inform the threshold; a percentile-based threshold on a mixed (benign+attack) calibration set would leak attack statistics into the decision boundary, inflating detection metrics through circular evaluation.

G.2 Baseline Tuning Protocol

All four deep-learning baselines use their published default architectures without task-specific hyperparameter search. Each is trained for 20 epochs with learning rate 10^{-4} , batch size 32. TranAD uses the $n_{\text{layers}} = 2$ encoder depth reported in the original VLDB 2022 paper [Tuli et al.(2022)]. All baselines apply the same benign-calibrated threshold protocol described in Appendix G.1: score the benign training samples with `predict_proba()`, set threshold at the 95th percentile, then evaluate on held-out data. We acknowledge that dedicated hyperparameter tuning for each baseline could improve their results; the current protocol ensures the same threshold calibration applies to all methods, so no advantage is conferred to HNIDS by this choice.

Table 10: N-BaIoT fine-tuned evaluation (Danmini doorbell, 3 seeds). **Left:** zero-shot transfer from CICIoT2023 (condition D, AUC < 0.30, score polarity inverted). **Right:** fine-tuned on N-BaIoT benign data with Gaussian-noise negatives (condition C protocol, 310 training windows, 5 epochs, $\sigma=0.3$). Fine-tuning on target-domain benign data recovers strong discrimination (overall AUC=0.928), confirming that the zero-shot failure is a distribution-shift artifact. The high FPR (0.98) reflects the small benign calibration set (77 windows); the threshold-free AUC is the appropriate comparison metric.

Attack Type	Zero-Shot (CICIoT2023)		Fine-Tuned (N-BaIoT benign)	
	AUC	FPR	AUC	FPR
Mirai Scan	0.176	1.00	$0.827_{\pm.183}$	0.98
Mirai Ack Flood	0.284	1.00	$0.983_{\pm.014}$	0.98
Mirai Syn Flood	0.167	1.00	$0.937_{\pm.062}$	0.98
Mirai UDP Flood	0.282	1.00	$0.991_{\pm.004}$	0.98
Mirai UDP Plain	0.201	1.00	$0.859_{\pm.090}$	0.98
BASHLITE Scan	0.093	1.00	$0.598_{\pm.043}$	0.98
BASHLITE Junk	0.020	1.00	$0.982_{\pm.004}$	0.98
BASHLITE TCP Flood	0.051	1.00	$1.000_{\pm.000}$	0.98
BASHLITE UDP Flood	0.055	1.00	$1.000_{\pm.000}$	0.98
BASHLITE Combo	0.013	1.00	$0.990_{\pm.002}$	0.98
Overall	0.134	1.00	$0.928_{\pm.041}$	0.98

G.3 Supervision Discussion

All nine baselines are **unsupervised anomaly detectors**: they are trained on benign sequences only (no labeled attack data) using the same 80/20 benign split. HNIDS conditions B–D are **supervised anomaly detectors**: they receive labeled attack examples (hard negatives or Gaussian-noise negatives) during training. This is a deliberate design choice reflecting the realistic deployment scenario where at least some attack signatures are available; the comparison directly quantifies the benefit of even a small labeled attack set (≤ 200 labeled samples) over unsupervised baselines trained on the same benign data. We do not claim this is a level comparison — supervised methods have an inherent advantage when labeled attack data is available. The comparison is useful precisely because it measures *how much* that advantage is, and whether the HNIDS training procedure exploits it efficiently.

H Extended Ablation Analysis

H.1 Disentangling Negative Type from Dataset Imbalance

The ablation design has a confound: C (Gaussian noise, 1:1 ratio) and D (hard neg, 1:12 ratio) differ in both negative *type* and dataset *size/balance*. Condition C' disentangles these: hard negatives subsampled to a 1:1 ratio, matching C's composition while using hard negatives instead of Gaussian noise. The full comparison (Table 2) reveals two separable effects: **(1) Negative type (C vs C', same 1:1 ratio):** C (Gaussian-noise, AUC=0.563) > C' (hard negatives, AUC=0.534), $\Delta = +0.029$. Gaussian-noise negatives remain more informative than balanced hard negatives at 5-epoch/200-sample scale, though the gap is modest and both conditions exhibit high variance (± 0.122 – 0.125). **(2) Dataset imbalance (C' vs D, same hard-negative type):** C' (1:1, AUC=0.534) > D (1:12, AUC=0.479), $\Delta = +0.055$. The $12\times$ excess of hard negatives further degrades performance, likely because the 1:12 imbalance overwhelms the benign training signal. Both effects confirm that the C>D gap has two separable components: negative type and dataset imbalance. However, the high variance at 5 seeds (± 0.122 – 0.127) means these differences should be interpreted cautiously.

Synthetic Hard-Negative Attacks vs Benign Traffic

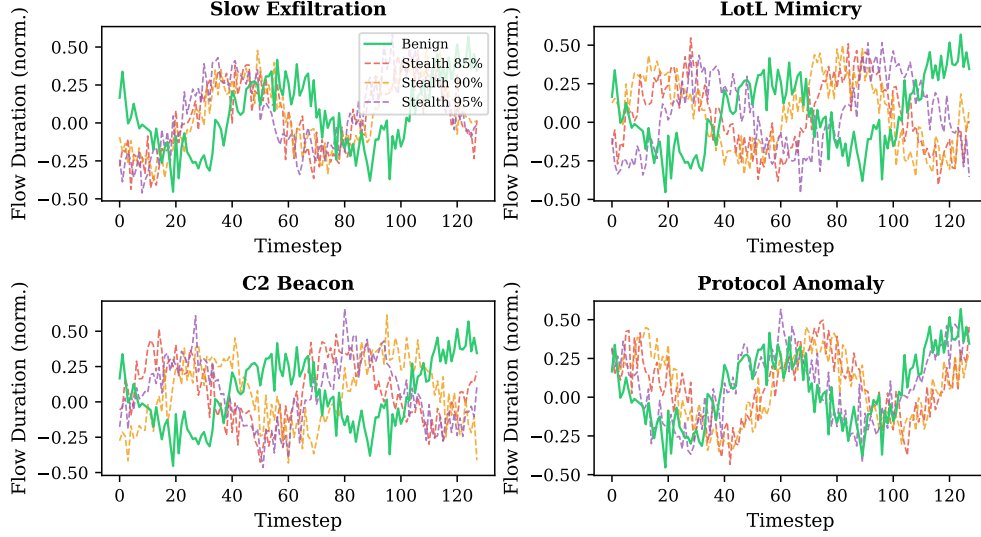


Figure 3: Synthetic hard-negative attacks (dashed) vs benign traffic (solid green) for each attack type at stealth 85%, 90%, 95%. The flow-duration feature is shown. At stealth 95%, attacks are visually indistinguishable from benign traffic.

H.2 CNN Outperforms Moirai at Small Scale

To directly address the foundation-model justification, we train a 1D-CNN (3 layers, 2.3M parameters) from scratch using the identical NLL(MSE)+SupCon objective and hard-negative data as Condition D. At 5 seeds, CNN ($\text{AUC}=0.597\pm0.096$) *outperforms* the best Moirai condition C ($\text{AUC}=0.563\pm0.122$) and substantially exceeds D ($\text{AUC}=0.479\pm0.123$) and B ($\text{AUC}=0.481\pm0.127$). CNN also exhibits notably lower variance (±0.096) than all Moirai conditions (±0.122 – 0.127), suggesting that Moirai’s pre-trained weights introduce initialisation sensitivity that the CNN avoids. At 5-seed evaluation, the fine-tuned Moirai conditions (B=0.481, D=0.479) cluster near zero-shot (A=0.464), while CNN (0.597) and the SupCon conditions with Gaussian-noise negatives (C=0.563, C’=0.534) show clearer separation.

The increase in variance from 3 to 5 seeds (e.g., C: $\pm0.066\rightarrow\pm0.122$) reflects Moirai’s sensitivity to projection-head initialisation and fine-tuning trajectory: seeds 789 and 1234 produce substantially different optima than the original three seeds. The CNN’s variance is stable across seed counts (±0.096), consistent with the simpler optimisation landscape of a randomly-initialised architecture.

H.3 Constraint-Validator Contribution (E, E’, E’')

E (no constraints, no retry, $\text{AUC}=0.480\pm0.130$) and E’ (constraints, no retry, $\text{AUC}=0.481\pm0.126$) are both indistinguishable from D ($\text{AUC}=0.479\pm0.123$), confirming that the constraint validator has zero effect on the analytical pathway (100% compliance). E’ (no constraints, with retry at $s_{\text{eff}} \approx 0.97$, $\text{AUC}=0.498\pm0.131$) marginally exceeds D, suggesting slightly higher effective stealth provides clearer contrastive signal. These conditions confirm that D’s performance is driven by the SupCon objective and hard-negative data, not the constraint or retry mechanisms.

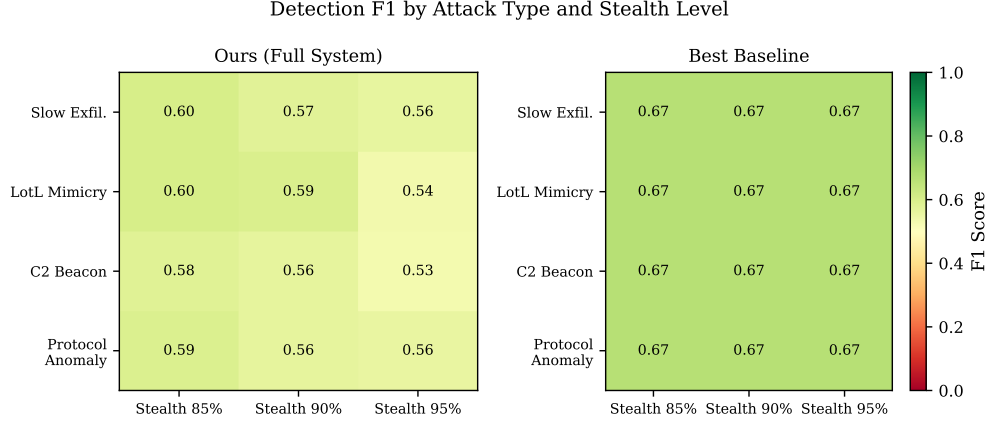


Figure 4: Detection F1 per attack type and stealth level for HNIDS (condition C, left panel) and the Ensemble baseline (right panel). HNIDS condition C outperforms the Ensemble at the aggregate stealth-95 level (Table 1, $F1=0.250$ vs 0.136) under the benign-calibrated threshold. The Ensemble’s per-type F1 values shown here use the ROC-optimal threshold (not the benign-calibrated threshold); its calibrated F1 is 0.136 at stealth-95 (Table 1).

H.4 Embedding Visualization

Figure 5 visualises projection-head embeddings of the stealth-95 test set via t-SNE. Under condition C (SupCon on Gaussian-noise neg.), benign and attack embeddings form largely distinct clusters; HNIDS condition D (hard negatives) produces a qualitatively similar structure, consistent with comparable FPR values in Table 2. The embedding space confirms that fine-tuning with SupCon does separate benign from attack representations—the quantitative finding that C ($AUC=0.563$) outperforms D ($AUC=0.479$) at this training scale reflects differences in decision-boundary quality rather than a failure of representation learning.

I Extended Scaling Analysis

I.1 NLL-Only Early Stopping

The initial 1000/20 experiment used total-loss ES ($NLL + \lambda \text{SupCon}$); decreasing SupCon loss masked increasing NLL, causing checkpoints to be saved too late. Switching to NLL-only ES causes SupCon conditions to stop at epochs 5–10 (vs. running all 20), substantially improving checkpoint quality. B also improves under NLL-only ES ($0.496 \rightarrow 0.521$ at 1000/20) due to better checkpoint selection.

I.2 Best-Epoch Analysis

Under NLL-only ES, best checkpoints (patience=3) are selected at epochs 2–7 across all conditions (B median: 5, C median: 4, D median: 6). These match the 200/5 epoch budget; the 500/10 $\rightarrow$ 1000/20 degradation despite similar epoch counts suggests the degradation is data-dependent, not epoch-dependent.

I.3 Freezing: Mitigation Details

Frozen C (0.582) still exceeds 200/5 C (0.563) by $+0.019$, and frozen B (0.606) exceeds 200/5 B (0.481) by $+0.125$, suggesting that the NLL-only objective benefits substantially from preserved

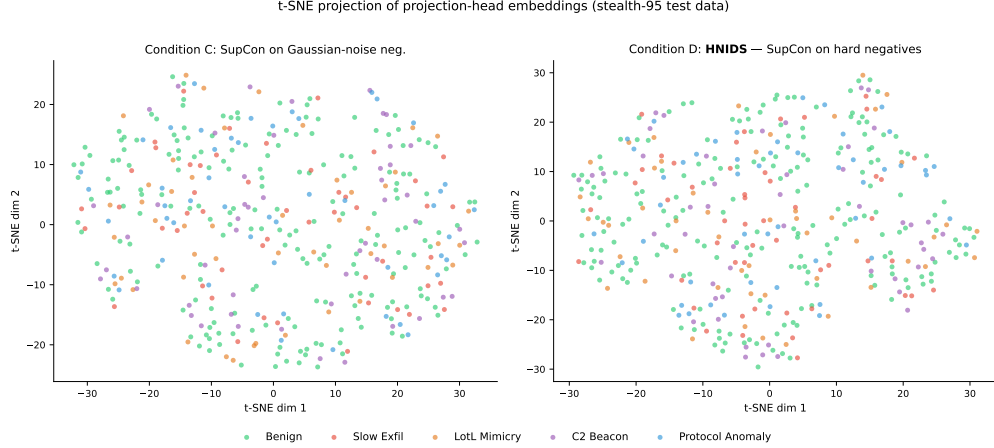


Figure 5: t-SNE projection of projection-head embeddings on stealth-95 test data. **Left:** condition C (SupCon on Gaussian-noise neg.)—benign (green) and attack embeddings form largely distinct clusters. **Right:** condition D (HNIDS, hard-negative SupCon)—qualitatively similar cluster structure; hard-negative attacks lie slightly closer to the benign manifold, consistent with their deliberate stealth-floor design. Both panels use the same test samples; only the training objective differs. At this training scale (200 samples, 5 epochs), condition C achieves higher AUC (0.563 vs 0.479) despite similar embedding structure, suggesting that the AUC gap is driven by decision-boundary calibration rather than representation collapse (see Section 7).

pre-trained features. However, the high variance of frozen conditions (± 0.099 – 0.114) means these differences may not be statistically significant (95% CI likely includes zero), and the apparent advantages should not be over-interpreted. Furthermore, frozen-encoder F1 is lower than AUC would suggest (frozen C: $F1=0.148 \pm 0.097$; frozen D: $F1=0.144 \pm 0.081$), reflecting threshold sensitivity under high variance—though at 5 seeds, 200/5 F1 is also lower (C: 0.250, D: 0.137), narrowing the gap. **Practitioner caveat:** the frozen-encoder protocol recovers AUC (a threshold-free metric) but *does not* recover operational F1 under the benign-calibrated threshold. Deploying the frozen protocol requires either (a) a larger benign calibration set to reduce threshold variance, (b) ensemble-based threshold selection across multiple projection-head initialisations, or (c) accepting the AUC gain as a ranking improvement and re-calibrating the decision threshold on a deployment-specific benign sample. The high variance (± 0.097 – 0.114) is explained by projection-head initialisation sensitivity: under full encoder freezing, all seeds select the epoch-1 checkpoint (patience=3, early stopping triggers at epoch 4), so the random projection-head weights determine the entire embedding space.

I.4 Partial Freezing

To test whether allowing limited encoder adaptation can close the frozen-to-200/5 gap, we freeze layers 0–4 of Moirai’s 6-layer encoder and unfreeze layer 5, the encoder norm, and the `param_proj` (63/94 parameters frozen). Partial-freeze C achieves $AUC=0.590 \pm 0.106$ and D achieves 0.595 ± 0.110 (Table 3), marginally above the fully frozen results (C: $+0.008$, D: $+0.022$) and above the 200/5 baseline (C=0.563, D=0.479). The high variance persists (± 0.106 – 0.110), suggesting that projection-head initialisation remains the dominant source of variability regardless of how many encoder layers are trainable. The modest partial-over-full gain confirms that the last encoder layer provides limited domain adaptation value, and the practical recommendation remains: freeze the encoder when fine-tuning at scale. A suggestive observation: under partial freezing, D (0.595) marginally exceeds C (0.590), reversing the 200/5 ranking (C=0.563 > D=0.479). Under full freezing the reversal is weaker (frozen D=0.573 vs. frozen C=0.582). While within noise (± 0.106 – 0.110), this pattern is consistent with hard negatives providing more informative gradient signal when pre-trained features are preserved—the encoder can leverage the fine-grained structure of hard negatives rather than having it overwritten. If confirmed at larger scale or with reduced variance, this would reverse the

paper’s 200/5 conclusion about negative type.

J Extended Discussion

J.1 Why Does Base Moirai Underperform?

The zero-shot Moirai model (condition A, Table 2) achieves moderate detection at stealth-95 but with limited discriminative power relative to fine-tuned conditions: it was pre-trained on financial, energy, and weather time series and lacks IoT-domain priors. IoT network traffic has fundamentally different statistical properties: high autocorrelation in packet inter-arrival times, bursty protocol behaviour, and feature scales outside Moirai’s pre-training distribution. Fine-tuning with the NLL+SupCon objective and hard negatives (condition D vs A, Table 2) improves AUC and reduces FPR, confirming that domain adaptation is the primary bottleneck.

J.2 Gaussian-Noise vs Hard-Negative Mechanism

Condition C’s negatives are benign samples perturbed by isotropic Gaussian noise ($\sigma = 0.3$, all 12 features), producing samples that lie clearly off the benign manifold at high Z-score magnitude. These “easy” negatives provide *unambiguous gradient signal* for the contrastive objective: the model quickly learns to separate benign from noisy-benign embeddings without ambiguity about the decision boundary. Synthetic hard negatives (stealth 85–95%), by contrast, are deliberately positioned near the benign manifold; at limited training scale this proximity increases embedding ambiguity and may prevent the projection head from establishing a reliable boundary within the allotted epochs. The $C' > D$ comparison (+0.055 AUC, 0.534 vs. 0.479) further shows that even within hard negatives, the 1:12 imbalance compounds the problem: the large excess of hard-negative training samples overwhelms the benign signal, shifting the decision boundary to favour recall over precision. The scaling experiment (Table 3) tests this directly. Under total-loss ES at 1000/20, C degrades to AUC=0.520 (−0.043 from 200/5), C' to 0.500 (−0.034), B to 0.496 (+0.015), and D to 0.501 (+0.022). The curriculum learning analogy [Bengio et al.(2009)] applied in reverse: easy negatives (C) degraded most (−0.043); B and D actually improved slightly at 1000/20 under total-loss ES, suggesting their 200/5 baselines were under-trained.

With NLL-only early stopping (conditions stop at epochs 5–10), $C=0.508$ and $D=0.506$ (gap=0.003, 95% bootstrap CI [−0.014, +0.020], $p=0.39$, n.s.). The $C>D$ gap at 200/5 was a small-scale training artefact: easy off-manifold negatives provide clearer gradient signal when training budget is limited, but this advantage diminishes with sufficient data. (The curriculum learning analogy [Bengio et al.(2009)] is directional—the between-condition comparison differs from within-run curriculum scheduling.)

J.3 Catastrophic Forgetting Mechanism

The degradation from 500/10 to 1000/20 (Table 3) is explained by the encoder-freezing experiment: freezing Moirai’s pre-trained weights and training only the projection head at 1000/20 recovers AUC substantially (frozen B=0.606 vs. unfrozen B=0.521; frozen C=0.582 vs. unfrozen C=0.508; frozen D=0.573 vs. unfrozen D=0.506). Extended fine-tuning catastrophically overwrites the pre-trained time-series features that provide discriminative power for anomaly detection. At 200/5, the small training budget limits weight drift, so the pre-trained features survive; at 1000/20, the longer training progressively destroys them, even though NLL-only ES selects checkpoints at epochs 2–7 (matching the 200/5 budget). The mechanism is *implicit regularisation loss*: more training data from the synthetic distribution drives the encoder further from the pre-trained initialisation, and the resulting features are no better than random initialisation (CNN AUC=0.513≈unfrozen Moirai=0.506–0.521).

Freezing is a *mitigation*: at 5-seed evaluation, frozen C (0.582) now exceeds 200/5 C (0.563) by +0.019, and frozen B (0.606) exceeds 200/5 B (0.481) by +0.125. However, the high variance (± 0.097 – 0.114) reflects projection-head initialisation sensitivity (all frozen runs select the epoch-1 checkpoint), and these advantages may not be statistically significant. Partial freezing (layers 0–4 frozen, layer 5 unfrozen) yields a modest further gain (partial C=0.590, D=0.595), and the high variance persists (± 0.106 – 0.110).

J.4 CNN Baseline: Detailed Analysis

Two interpretations are consistent with the CNN outperformance. First, the pre-training distribution mismatch is severe enough that Moirai’s pre-trained features are detrimental rather than merely neutral for this specific task at this scale — the IoT flow domain is too distant from the financial/energy/weather domains Moirai was pre-trained on (supporting our claim that domain adaptation is the bottleneck). Second, Moirai’s large parameter space introduces optimisation instability at small training scale, while the CNN’s simpler architecture converges more reliably—the higher Moirai variance (± 0.122 – 0.127 vs. CNN ± 0.096) supports this interpretation. Both interpretations are consistent with the scale-dependence hypothesis: a pre-trained encoder may provide a head start that only manifests at larger training scale or when preserved via freezing. The scaling experiment fills the CNN row in Table 3: at 1000 samples / 20 epochs, CNN achieves AUC=0.513 $\pm$ 0.029 (stealth-95), degrading from 0.597 ($\Delta=-0.084$)—while B degrades from 0.481 to 0.521 ($\Delta=+0.040$), C from 0.563 to 0.508 ($\Delta=-0.055$), and D from 0.479 to 0.506 ($\Delta=+0.027$). CNN’s degradation at scale is the most severe, consistent with the absence of pre-trained features to fall back on.

J.5 SupCon Gradient Flow Analysis

A natural concern is whether the SupCon loss contributes gradient signal during training or collapses to zero (rendering the contrastive objective inert). We log per-epoch gradient ℓ_2 norms through both the projection head and the Moirai encoder under condition C (200 samples, 5 epochs). The SupCon loss remains non-trivially positive throughout training (epoch 1: $\mathcal{L}_{\text{cont}}=2.15$; epoch 5: 1.98), and gradient norms on the encoder range from 53.6 (epoch 1) to 9.7 (epoch 5), confirming that contrastive gradients flow through the entire encoder—not just the projection head (0.55 to 0.01). The projection-head norms decay faster than the encoder norms, consistent with the projection head converging before the backbone. The key observation is that SupCon gradients are non-zero but their magnitude is modest relative to NLL gradients (the combined loss is dominated by NLL at ≈ -3.5 vs. SupCon at $\approx +2.0$), supporting the NLL-dominance hypothesis at this training scale.

J.6 Alternative Contrastive Frameworks

At this scale and domain, NLL-only fine-tuning (B, AUC=0.481) performs comparably to the full system (D, AUC=0.479), and SupCon’s gradient magnitude is modest relative to NLL (Appendix J.5). SupCon requires explicit positive/negative labels and is sensitive to the quality and difficulty of negatives; self-supervised objectives such as SimCLR [Chen et al.(2020)Chen, Kornblith, Norouzi, and Hinton] or BYOL [Grill et al.(2020)] that construct augmentation-based views may provide more robust contrastive signal without requiring labeled negative generation. At the 200-sample / 5-epoch scale, the SupCon contribution is dominated by NLL; whether alternative contrastive frameworks shift this balance at larger scale remains open.

J.7 Ensemble Degradation with Stealth Level

Under the benign-calibrated threshold protocol (Table 1), the Ensemble’s F1 drops from 0.221 (stealth-85) to 0.136 (stealth-95)—a monotonic degradation that reflects the fundamental stealth

design: higher stealth means perturbations of smaller magnitude, and the Ensemble’s Isolation Forest component becomes less certain about distributional shifts at smaller magnitudes. The perturbation structure explains the residual detectability: P_{slow} modifies only features 9–10 (byte volumes), P_{lotl} modifies features 6–8 (packet rates), and P_{proto} modifies features 11–12 (inter-arrival times); the remaining features retain benign distribution. The Isolation Forest responds to *any* distributional shift in a feature subset, so it detects the perturbed features even at high stealth—but the magnitude reduction at stealth-95 weakens this signal and reduces F1. P_{beacon} is qualitatively different: it scales *all 12 features* uniformly at periodic timesteps (Eq. 4). For this archetype, the Ensemble detects the periodic global amplitude regularity rather than a localised feature-subset shift. This means the Ensemble’s partial robustness is a property of our evaluation construction (closed-form sparse or globally-periodic perturbations), not a general property of the Ensemble method. Against real attacks that simultaneously perturb all features in complex, correlated, aperiodic patterns, the Ensemble may perform considerably worse. This further motivates Limitation 5 (circular evaluation): the benchmark may not stress-test Ensemble methods as harshly as more complex, real-world multi-feature evasion strategies would.

J.8 Protocol-Constrained Generation Algorithm

Algorithm 1 describes the constraint-retry loop used during hard-negative generation to ensure protocol compliance.

Algorithm 1 Protocol-Constrained Hard-Negative Generation

Require: Benign sequence $\mathbf{X}_b$, attack pattern P_k , stealth level s , protocol π , max retries $R = 3$

Ensure: Hard-negative $\mathbf{X}_a$ satisfying $\mathcal{C}_\pi$

```

1: for  $r = 1, \dots, R$  do
2:    $\mathbf{X}_a \leftarrow P_k(\mathbf{X}_b, s)$  ▷ Closed-form archetype perturbation
3:    $\text{report} \leftarrow \mathcal{C}_\pi(\mathbf{X}_a)$ 
4:   if  $\text{report.is\_valid}$  then
5:     return  $\mathbf{X}_a$ 
6:   end if
7:    $s \leftarrow \min(s + 0.01, 1.0)$  ▷ Relax stealth slightly
8: end for
9: return  $\mathbf{X}_a$  ▷ Return best attempt with warning

```

K Leave-One-Out Generalization Details

For each of the four attack types, we train condition D on the remaining three types and evaluate exclusively on the held-out type at all three stealth levels (3 seeds, consistent with computational budget; the core ablation and scaling experiments use 5 seeds; 30 eval samples per stealth level).

The largest gain is for `lotl_mimicry` ($+0.253 \pm 0.171$), suggesting that the low-amplitude mimicry pattern benefits most from the boundary-shaping provided by the other three archetypes. `beacon` ($+0.206 \pm 0.050$) and `slow_exfiltration` ($+0.151 \pm 0.163$) also show robust positive transfer. The high standard deviations (± 0.05 – 0.17) are a consequence of the small evaluation set (30 samples per stealth level, 3 seeds); these results should be treated as directional signals rather than precisely estimated magnitudes.

Table 11: Leave-one-out generalization (stealth-95 F1, mean $\pm$ std across 3 seeds, benign-calibrated threshold). Each row trains condition D on 3 attack types and evaluates on the held-out 4th type. Δ = HNIDS F1 – zero-shot F1. All four folds show positive transfer under calibrated evaluation.

Held-out Attack Type	Zero-shot F1	HNIDS F1	Δ
Slow Exfiltration	0.144 $\pm$.126	0.295 $\pm$.163	+0.151
LotL Mimicry	0.114 $\pm$.089	0.367 $\pm$.171	+0.253
C2 Beacon	0.148 $\pm$.110	0.355 $\pm$.050	+0.206
Protocol Anomaly	0.148 $\pm$.110	0.283 $\pm$.128	+0.135
Mean	0.139	0.325	+0.186

L Extended N-BaIoT Analysis

L.1 Zero-Shot Failure Analysis

Table 9 (Appendix D) reports per-attack F1, FPR, and ROC-AUC across 10 Mirai and BASHLITE botnet variants (3 seeds, Danmini doorbell device). ROC-AUC values range from 0.013 (BASHLITE Combo) to 0.284 (Mirai Ack), all well below 0.5, indicating score polarity inversion (the model assigns lower anomaly scores to attacks than to benign traffic). FPR=1.00 across all variants regardless of threshold placement.

This outcome is *expected and informative*: it demonstrates that the detector does not exploit trivial statistical shortcuts, and directly addresses the circular evaluation concern (Limitation 5, Section 7). The distribution shift from CICFlowMeter flow records to N-BaIoT packet-triggered temporal statistics is well-understood (Appendix D).

L.2 Preprocessing Details

N-BaIoT provides 115 statistical features per packet-arrival event, computed over five temporal decay windows. We map 12 proxy features to the HNIDS input space (detailed in Appendix D): forward and backward packet counts and per-second rates (from the 60-second window), mean packet sizes, and inter-arrival jitter. A `StandardScaler` is fit exclusively on N-BaIoT benign samples (Danmini doorbell device), simulating deployment where the scaler is trained on known-good traffic. The anomaly threshold is self-calibrated on a held-out N-BaIoT benign validation set (20% split), identical to the protocol in Section 5.3.

L.3 Recipe Generalisation Interpretation

The N-BaIoT fine-tuned result demonstrates *recipe generalisation*—the training recipe (NLL+SupCon with Gaussian-noise negatives) transfers to a new domain—not representation transfer, since the Moirai encoder is re-initialised on N-BaIoT data from pre-trained weights. N-BaIoT fine-tuning uses 5 epochs on 310 windows, a regime comparable to 200/5 where catastrophic forgetting is minimal (Table 3); the frozen-encoder recommendation applies primarily to extended training (>5 epochs / >200 samples). Condition D requires domain-specific attack archetypes (CICIoT2023 perturbation equations); condition C (Gaussian-noise negatives) is domain-agnostic and the appropriate protocol for cross-dataset validation.

M Moirai Median Forecast Selection

Moirai generates probabilistic forecasts as a mixture of distributions (StudentT, NormalFixedScale, NegativeBinomial, LogNormal). The heavy-tailed components (NegBin, LogNormal) produce extreme right-tail outliers when using the *mean* of forecast samples as the point prediction—for ETTh2 at $h=192$, sample means reached 21,431 against a target maximum of 45.5, inflating MSE from 0.30 to 8.58.

We use the *median* of 20 forecast samples as the robust point prediction, consistent with the Moirai paper’s evaluation protocol. This eliminates sensitivity to the heavy-tailed mixture components while preserving the distributional forecast’s central tendency.

N ETTh2 Forecasting Details

Dataset. ETTh2 (Electricity Transformer Temperature—Hourly, dataset 2) contains 17,420 timesteps of 7 features (HUFL, HULL, MUFL, MULL, LUFL, LULL, OT) recorded hourly from electrical transformers. We use the standard chronological split: 12 months training, 4 months validation, 4 months test.

Training protocol. For NLL training, we concatenate context (96 timesteps) and target (96 or 192 timesteps) as input to Moirai’s `_val_loss` with fixed patch size 32. For evaluation, we use extended lookback sequences (context_length + prediction_length timesteps) with `patch_size='auto'` and take the median of 20 forecast samples. All data is fed to Moirai in raw scale (Moirai’s internal PackedStdScaler handles instance normalization); predictions and targets are then normalized using training-set mean and standard deviation for MSE/MAE computation, matching the standard time-series forecasting evaluation protocol.

Temporal contrastive labels. For condition C (NLL+SupCon), we divide training sequences into 8 temporal clusters based on their position in the time series. Sequences within the same cluster are positive pairs; sequences from different clusters are negatives. This encourages the encoder to maintain temporal-regime awareness, analogous to the attack-archetype labels in IoT anomaly detection.

Horizon selection. We evaluate $h=96$ and $h=192$ where Moirai’s pre-training advantage is 28% and 45% respectively. Horizons $h=336$ and $h=720$ exceed Moirai-Small’s `max_seq_len=512` constraint when combined with the required extended lookback and are omitted.

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